

Basic Automobile Engineering

Course 1051 · Semester I · A first-principles guide to vehicle architecture, thermodynamics, engines, fuels and safe workshop observation.

Programme	Automobile Engineering
Course type	Practicum
Teaching scheme	3:0:3:0
Contact hours	90
Credits	4.5
CIE / ESE	40 / 100

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official SITTR Kerala **Diploma Curriculum Revision 2026** syllabus PDF for course **1051 Basic Automobile Engineering**. No Revision 2021 subject content has been merged.

Verified official-source inconsistency: Page 1 lists CIE Marks 40 and ESE Marks 100. The detailed assessment table on page 2 splits theory as 20 CIE + 60 ESE and practical as 20 CIE + 40 ESE, which totals the same 40 CIE and 100 ESE. This handbook therefore displays both the overall total and the official theory/practical split rather than treating them as contradictory totals.

Not specified in the supplied PDF: official textbooks, reference books, websites, detailed equipment list and an official model question paper.

Course Dashboard

90 h

45 theory + 45 practical

4.5

credits

course outcomes

140

total assessment marks

Official teaching and assessment scheme

L	T	P	S	Total h	Theory CIE/ESE	Practical CIE/ESE
3	0	3	0	90	20 / 60	20 / 40

Prerequisite: Basic knowledge in science at Secondary Education level.

How to Use This Handbook

1. Read the concept and identify the physical component or process.
2. Study the diagram and Malayalam support.
3. Follow the demonstration or worked procedure.
4. Attempt the self-check and expected questions before opening the answer.
5. Use the Procedure Bank and Quick Revision sections before practical and theory examinations.

Objectives and Course Outcomes

Official objectives

1. Describe automobile evolution, industry players and physical architecture.
2. Explain thermodynamic laws and cycles related to energy conversion.
3. Identify IC-engine components and operating principles.
4. Outline fuel properties, safety standards and storage requirements.

Official Course Outcomes

CO1: Describe historical development, major Indian manufacturers, chassis layouts and dimensions.**CO2:** Explain thermodynamic systems, laws and theoretical engine cycles.**CO3:** Differentiate engine components and 2-stroke, 4-stroke, petrol and diesel operation.

Curriculum Coverage Map and Learning Roadmap

Module	Official topics	L+P hours	CO	Handbook section
I	Evolution, manufacturers, chassis/body, layouts, dimensions, weights, engine mounts	10+10	CO1	Module I
II	Systems, properties, laws, processes, Carnot/Otto/Diesel cycles	12+12	CO2	Module II
III	Engine components, terminology, 4-stroke, 2-stroke, SI/CI comparison	12+12	CO3	Module III
IV	Refining, storage, fuel properties, BS-VI, biofuels, filling station	11+11	CO4	Module IV

Vehicle

Thermodynamics

Engines

Fuels

Practical Centre

Questions

Quick Revision

Module I – Introduction to Automobile (10 L + 10 P)

An automobile is an integrated system in which structure, power source, transmission, steering, suspension, brakes and controls work together to transport people or goods safely.

Evolution and Indian manufacturers

Development progressed from steam vehicles to petrol/diesel IC-engine vehicles, mass production, electronic control, hybridisation and electric mobility. In India, the syllabus highlights Hero and Bajaj (2W/3W), Maruti Suzuki and Tata Motors (passenger/commercial), Ashok Leyland (heavy commercial) and Mahindra (utility, tractor and commercial products). Plant locations and product ranges must be checked against current manufacturer information during class because they can change.

Malayalam support: Automobile evolution എന്നത് engine മാത്രം മാറിയ കഥയല്ല. Manufacturing, safety, electronics, emission control, materials, fuel and user needs എല്ലാം ചേർന്നുള്ള മാറ്റമാണ്.

Chassis, body and vehicle layouts

Chassis is the load-carrying framework with major running units. **Body** encloses occupants or load. A conventional HMV commonly uses a separate frame; many passenger cars use monocoque/unibody construction. Layout is described by engine position and driven wheels: front-engine rear-wheel drive, front-engine front-wheel drive, rear-engine rear-wheel drive and all-wheel drive.

Power flow example: Engine → clutch/torque converter → gearbox → propeller shaft or transaxle → differential → axle shafts → wheels.

Dimensions, weights and centre of gravity

- **Wheelbase:** distance between front and rear axle centre lines.
- **Wheel track:** distance between left and right wheel centre lines on the same axle.
- **Overhang:** vehicle length beyond the nearest axle centre.
- **Ground clearance:** minimum distance between ground and lowest specified vehicle point.
- **Kerb weight:** vehicle ready for normal road use without payload.
- **GVW:** maximum permitted total operating weight.

A lower centre of gravity generally improves resistance to rollover, but suspension travel, ground clearance and load distribution also matter.

Common error: Wheelbase is not total vehicle length; wheel track is not tyre width.

Engine mounts

Mounts locate the powertrain, carry static and dynamic loads, isolate vibration and control torque reaction. Rigid mounts transmit vibration; rubber/flexible mounts isolate better; hydraulic mounts use fluid damping for wider-frequency control. Selection depends on engine mass, excitation frequency, torque, packaging, durability, heat and chemical exposure.

Malayalam support: Engine mount loose ആണെങ്കിൽ vibration കൂടും. അതിയായി soft mount ഉപയോഗിച്ചാൽ powertrain movement കൂടും. Isolation-നും control-നും ഇടയിലെ balance ആണ് selection.

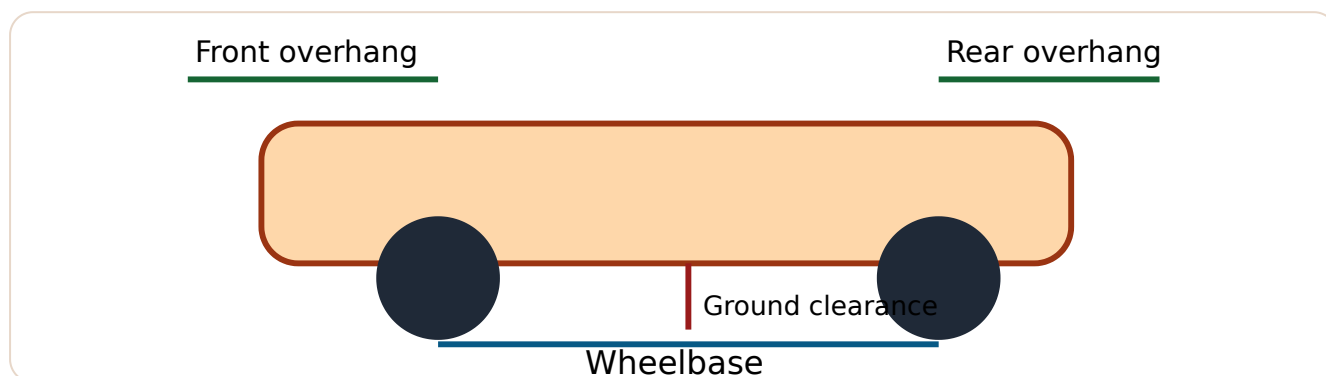


Figure 1. Key longitudinal vehicle dimensions; always measure on a level surface using the defined reference points.

Module II — Basics of Thermodynamics (12 L + 12 P)

System, surroundings, boundary and universe

A thermodynamic **system** is the chosen matter or region. Everything outside is the surroundings; the separating surface is the boundary. Open systems exchange mass and energy (engine cylinder during intake/exhaust, radiator coolant passage). Closed systems exchange energy but not mass for the selected interval (sealed gas in a piston-cylinder). An isolated system ideally exchanges neither.

Malayalam support: System type component name നോക്കി തീരുമാനിക്കരുത്. പഠിക്കുന്ന time interval-ലും chosen boundary-യിലും mass കടക്കുമോ എന്ന് ആദ്യം നോക്കണം.

Properties and equilibrium

Intensive properties do not depend on amount: temperature, pressure, density, viscosity. **Extensive** properties depend on amount: mass, volume, total internal energy. Equilibrium means no unbalanced thermal, mechanical or chemical tendency within the system.

Laws of thermodynamics

Zeroth: establishes temperature through thermal equilibrium. **First:** energy is conserved.

For a closed system: $Q - W = \Delta U$. For steady-flow devices: heat + inlet energy = work + outlet energy.

Second: heat naturally flows from high to low temperature and no heat engine can convert all supplied heat into work. **Third:** entropy approaches a minimum as temperature approaches absolute zero; absolute zero cannot be reached by a finite process.

Engineering implication: An engine requires radiator, exhaust and lubrication because rejected heat and friction cannot be eliminated.

Processes and ideal cycles

Isochoric: constant volume.

Isobaric: constant pressure.

Isothermal: constant temperature.

Adiabatic: no heat transfer.

Otto cycle: ideal SI-engine cycle with constant-volume heat addition.

Diesel cycle: ideal CI-engine cycle with constant-pressure heat addition.

Carnot cycle: ideal reversible benchmark, not a practical vehicle engine cycle.

Worked energy example: An engine receives 1200 kJ heat and delivers 360 kJ work per cycle. Heat rejected = $1200 - 360 = 840$ kJ. Thermal efficiency = $360/1200 \times 100 = 30\%$. The remaining energy is not “lost”; it leaves mainly through exhaust, coolant and friction-related heating.

Module III — Engine Basics (12 L + 12 P)

Fixed parts

Cylinder block, cylinder head, crankcase and manifolds provide structure and passages.

Moving parts

Piston, rings, connecting rod, crankshaft, camshaft and valves transmit and control motion.

Support systems

Fuel, ignition/injection, lubrication, cooling, intake and exhaust enable controlled operation.

TDC and **BDC** are piston extreme positions. **Stroke** is the distance between them; **bore** is cylinder diameter. Swept volume is displaced by the piston; clearance volume remains above the piston at TDC.

Swept volume per cylinder, $V_s = (\pi/4)D^2L$. Compression ratio, $r = (V_s + V_c)/V_c$.

Example: $D = 80 \text{ mm}$, $L = 90 \text{ mm}$. $V_s = \pi/4 \times 0.08^2 \times 0.09 = 0.000452 \text{ m}^3 \approx \mathbf{452 \text{ cm}^3}$.

Four-stroke petrol and diesel operation

1. **Intake:** piston moves TDC→BDC. SI draws air-fuel mixture; CI draws air.
2. **Compression:** both valves closed; piston BDC→TDC.
3. **Power:** SI uses spark ignition; CI fuel is injected into hot compressed air and self-ignites.
4. **Exhaust:** exhaust valve opens; piston expels products.

Malayalam support: Petrol engine-ൽ spark plug ignition timing നിർണ്ണയം. Diesel engine-ൽ compression മൂലം air temperature ഉയരുകയും injector fuel spray നൽകുകയും ചെയ്യുന്നു.

Two-stroke petrol engine and scavenging

A two-stroke completes a cycle in one crankshaft revolution. Ports replace or supplement valves. Fresh charge entering through the transfer port pushes exhaust gas toward the exhaust port; this is scavenging. Simplicity and high power-to-weight are advantages, but short-circuiting of charge, lubrication and emissions are major limitations.

SI and CI engine comparison

Feature	SI / Petrol	CI / Diesel
Ignition	Spark plug	Self-ignition after injection
Charge during intake	Usually air-fuel mixture	Air only
Compression ratio	Generally lower	Generally higher
Fuel rating	Octane number	Cetane number
Typical traits	Smoother, lighter	Higher low-speed torque and efficiency

Module IV — Fuels (11 L + 11 P)

Origin, refining and fuel journey

Crude oil is extracted, transported to a refinery, separated by fractional distillation according to boiling range, further converted and treated, blended to specification, stored at terminals, transported by approved tankers and dispensed at retail outlets.

Flow: Crude extraction → transport → desalting → fractional distillation → conversion/treatment → blending and quality test → terminal storage → tanker → underground retail tank → dispenser → vehicle tank.

Storage, handling and filling-station safety

Fuel stations use underground tanks, fill points, vents, dispensers, emergency isolation, earthing/bonding, fire extinguishers and controlled traffic paths. The vehicle system includes cap, filler neck, tank, vent/EVAP arrangement, pump, lines and level sender.

Safety: No smoking or open flame. Switch off ignition. Avoid static discharge and spills. Do not taste or deliberately inhale fuel. Use institution-approved samples and PPE for demonstrations. Flash/fire-point tests must be performed only with approved apparatus and instructor supervision.

Fuel specifications and grades

- **Octane number:** resistance of petrol to knock.
- **Cetane number:** ignition quality of diesel; higher generally shortens ignition delay.
- **Volatility:** tendency to vaporise; affects starting, mixture formation and vapour lock.
- **Viscosity:** resistance to flow; important for diesel injection and lubrication.
- **Calorific value:** heat released per unit mass or volume.
- **Flash point:** lowest temperature at which vapour flashes momentarily; **fire point** is where sustained burning occurs.
- **Sulfur content:** controlled to reduce harmful emissions and protect after-treatment.

BS-VI is India's stringent vehicle-emission and fuel-quality stage. It requires low-sulfur fuel and compatible engine/after-treatment systems. Ethanol blending can reduce petroleum dependence and alter octane, vapour behaviour, material compatibility and water sensitivity; always follow the vehicle manufacturer's approved blend.

Malayalam support: BS-VI fuel മാത്രം നിറച്ചാൽ പഴയ vehicle സ്വയം BS-VI vehicle ആകില്ല. Engine design, ECU calibration, catalytic after-treatment, sensors എന്നിവയും emission compliance-ന്റെ ഭാഗമാണ്.

Practical / Activity Centre

The official syllabus lists 24 practical activities. The formats below cover all activities while leaving instrument-specific values to the institution-approved lab manual.

Module I activities

1. Identify chassis and body structures.
2. Analyse LMV layout and power flow.
3. Analyse HMV layout and aggregate arrangement.
4. Measure LMV wheelbase, track, overhang and clearance.
5. Repeat dimensional study for HMV.
6. Inspect rigid, flexible and hydraulic engine mounts.

Module II activities

1. Classify automobile thermodynamic systems.
2. Demonstrate Zeroth Law and thermal equilibrium.
3. Compare intensive/extensive engine-oil properties.
4. Verify First Law through an energy balance.
5. Demonstrate Second-Law heat rejection.
6. Discuss Third Law and absolute zero.

Module III activities

1. Identify workshop tools.
2. Identify major engine components.
3. Measure bore, stroke and clearance references.
4. Demonstrate four-stroke petrol cycle.
5. Demonstrate four-stroke diesel cycle.

Module IV activities

1. Map the fuel journey.
2. Conduct safe visual fuel identification.
3. Study vehicle tank and filling system.
4. Study petrol-pump safety equipment.
5. Demonstrate flash/fire point with approved apparatus.
6. Interpret BS-VI and fuel labels.

Standard practical record format

- 1 Aim and CO:** State the exact activity and mapped outcome.
- 2 Tools/apparatus:** Record actual institution-provided equipment.
- 3 Safety:** PPE, isolation, hot-surface, moving-part and fuel precautions.
- 4 Procedure:** Numbered actions; never operate an engine or fuel apparatus without instructor authorisation.
- 5 Observation:** Use a table with item, measured value, unit and remarks.
- 6 Result and inference:** State what was identified, measured or verified and note uncertainty/errors.

Observation item	Instrument/reference	Measured/identified value	Unit	Remarks

Procedure and Standards Bank

Dimension measurement

Level surface → correct tyre pressure → identify axle centres/reference points → use tape/plumb line → repeat measurement → record unit and uncertainty.

Component identification

Isolate equipment → observe location and connection → identify function → trace input/output → compare with drawing/manual → record condition.

Fuel safety

Approved sample only → ventilation → no ignition source → PPE → labelled container → spill control → correct waste handling.

Expected / Practice Questions with Answers

1. Differentiate wheelbase and wheel track.

Wheelbase is longitudinal distance between front and rear axle centre lines. Wheel track is lateral distance between left and right wheel centre lines on one axle.

2. Why are hydraulic engine mounts used?

They combine elastic support with fluid damping, reducing vibration across a wider frequency range while controlling powertrain movement.

3. Classify a radiator as a thermodynamic system.

With coolant flowing through the selected control volume it is an open system because mass and energy cross the boundary.

4. State the First Law for a closed system.

$Q - W = \Delta U$, using the convention that heat supplied to the system and work done by the system are positive.

5. Calculate swept volume for bore 70 mm and stroke 80 mm.

$$V = \pi/4 \times 0.07^2 \times 0.08 = 0.0003079 \text{ m}^3 \approx 308 \text{ cm}^3 \text{ per cylinder.}$$

6. Explain the main difference between petrol and diesel four-stroke power initiation.

A petrol/SI engine uses a spark plug to ignite a compressed mixture. A diesel/CI engine injects fuel into highly compressed hot air and it self-ignites.

7. What is scavenging?

The process of removing burnt gases and replacing them with fresh charge, especially important in two-stroke engines.

8. Distinguish octane and cetane numbers.

Octane indicates petrol knock resistance; cetane indicates diesel ignition quality and tendency for shorter ignition delay.

9. Why is flash point lower than fire point?

At flash point enough vapour exists for momentary ignition; sustained burning requires a higher temperature and continuing vapour generation.

10. Does BS-VI fuel alone make a vehicle BS-VI compliant?

No. Compliance also requires compatible engine design, calibration, sensors and emission after-treatment.

Practice Model Paper — not an official SITTTR model question paper

1. Define chassis, body, kerb weight and GVW.
2. Sketch and label wheelbase, track, overhang and ground clearance.
3. Explain all four thermodynamic laws with automobile examples.
4. Draw P-V cycle names and compare Otto and Diesel ideal heat addition.
5. Draw and explain a four-stroke diesel engine cycle.
6. Compare SI and CI engines.
7. Explain crude-oil-to-vehicle fuel flow and filling-station safety.
8. Discuss octane, cetane, flash point, fire point and BS-VI.

Quick Revision

M1

Vehicle = structure + powertrain + running systems. Know dimensions, layouts, weights and mounts.

M2

Define boundary first. First Law conserves energy; Second Law requires heat rejection.

M3

Know parts, TDC/BDC, bore/stroke, compression ratio and stroke sequence.

M4

Know refining flow, fuel ratings, safe handling, BS-VI and ethanol-blend compatibility.

Exam checklist

- Use correct units and labelled diagrams.
- Do not confuse system classification with component name.
- In engine answers, state valve/port condition, piston direction and ignition method.
- In fuel answers, separate property, safety significance and emission relevance.

Glossary

Term	Meaning
Chassis	Load-carrying vehicle framework and major running units.
Monocoque	Structure in which body shell carries major loads.
Scavenging	Replacement of exhaust gas with fresh charge.
Entropy	Thermodynamic property related to energy dispersal and process direction.
Knock	Abnormal rapid combustion causing pressure oscillations.
Octane number	Petrol resistance to knock.
Cetane number	Diesel ignition-quality rating.
GVW	Maximum permitted total operating vehicle weight.

References

Official references: Not specified in the supplied official syllabus PDF.

Supplementary study guidance: Use current manufacturer workshop manuals, institution-approved laboratory manuals and applicable Indian automotive safety/emission documents under instructor supervision.